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NAME	ROCKY BANTILAN ADAYA
PROJECT TITLE	Adaya IT Solution: Corporate Project Lifecycle & Workflow Management System
SUBJECT: CODE: TIME:	IT15/L Integrative Programming and Technologies 8441 10:00 AM – 12:00 PM
TOPIC (Type of Business Process)	#37 Collaborative and Workflow Transactions System
Products/Services	Corporate Enterprises, Project-Based Organizations, and HR/Project Management Services
Agile Model - SDLC	The Adaya IT Solution: Corporate Project Lifecycle & Workflow Management System is designed to improve corporate project workflows by automating proposal submission, task management, milestone tracking, document handling, HR/admin functions, and executive reporting. Requirement identification was based on analyzing end-to-end workflows, ensuring the system meets the needs of employees, managers, project managers, HR/admin, and executives. Core requirements include: • Submitting, reviewing, approving, rejecting, or returning project proposals for revision • Assigning projects to managers and distributing tasks to team members • Tracking task execution, milestones, and overall project progress • Managing employee accounts, roles, and access permissions • Centralized document management with version control • Generating actionable reports and dashboards for decision-making • Real-time notifications and integration with external APIs for workflow intelligence Functional requirements focus on delivering features that support the workflow: • Employees can create and submit proposals, upload supporting documents, and track their status • Department Heads can review proposals, approve, reject, or return them for revision • Project Managers can assign tasks, set milestones, monitor progress, and review task outputs • HR/Admin can manage user profiles, roles, workloads, and module access • Executives can view read-only dashboards and reports • The system provides centralized document storage, version control, search capabilities, and secure retrieval • Notifications and alerts through Twilio API, and integrations with Nager.Date and NewsAPI • Reporting and analytics dashboards using Chart.js and Power BI

Non-functional requirements ensure reliability, security, and usability:

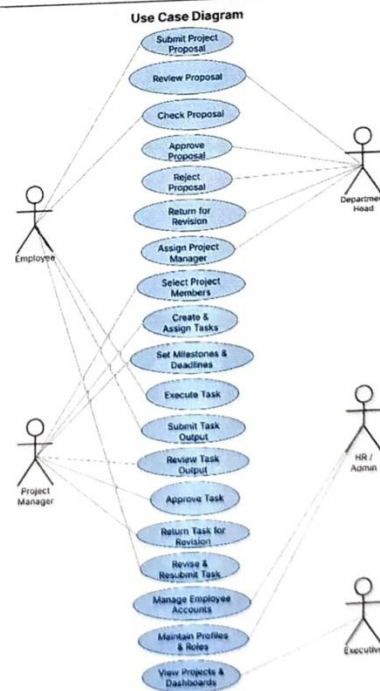
- Secure authentication and role-based access
- Data encryption, input validation, and secure API communication
- Logging, monitoring, and audit trails
- Scalable, user-friendly, and maintainable system
- Backup and recovery mechanisms for business continuity

The sprint planning strategy prioritizes development in phases:

- Early sprints focus on system setup, authentication, and proposal submission/review workflows
- Middle sprints implement task management, milestone tracking, document handling, and notifications
- Later sprints finalize reporting dashboards, HR/Admin modules, external API integrations, and performance/security optimizations

The planning outcome ensures that Adaya IT Solution: Corporate Project Lifecycle & Workflow Management System delivers a secure, efficient, and collaborative platform that automates project workflows, provides real-time updates, centralizes document management, and gives management actionable insights for decision-making.

Use Case Diagram
(Diagram Only)



Entity Relational Diagram

ACTIVITYLOGS			
int	LogID	PK	
int	UserID	FK	
string	Action		
string	Module		
datetime	Timestamp		

NOTIFICATIONS			
int	NotificationID	PK	
int	UserID	FK	
string	Message		
boolean	IsRead		
datetime	DateSent		

USERS			
int	UserID	PK	
string	Username		
string	Password		
string	Role		
boolean	IsActive		
datetime	DateCreated		

PROPOSALDOCUMENTS			
int	DocumentID	PK	
int	ProposalID	FK	
string	FileName		
string	FilePath		
datetime	UploadDate		

PROJECTPROPOSALS			
int	ProposalID	PK	
int	EmpID	FK	
string	Title		
string	Description		
string	Status		
datetime	DateSubmitted		

EMPLOYEES			
int	EmpID	PK	
int	UserID	FK	
string	EmployeeNumber		
string	LastName		
string	FirstName		
string	MiddleName		
string	Department		
string	Position		
string	Email		
date	DateHired		

PROJECTPROPOSALREVIEWS			
int	ProposalReviewID	PK	
int	ProposalID	FK	
int	ReviewedBy	FK	
string	ReviewStatus		
string	ReviewNotes		
datetime	ReviewDate		

PROJECTS			
int	ProjectID	PK	
int	ProposeID	FK	
int	ProjectManagerID	FK	
string	ProjectName		
date	StartDate		
date	EndDate		
string	Status		

MILESTONES			
int	MilestoneID	PK	
int	ProjectID	FK	
string	Title		
date	TargetDate		
string	Status		
datetime	LastUpdated		

PROJECTMEMBERS			
int	MemberID	PK	
int	ProjectID	FK	
int	EmpID	FK	
string	Role		
datetime	DateAssigned		

TASKS			
int	TaskID	PK	
int	ProjectID	FK	
int	MilestoneID	FK	
int	MemberID	FK	
string	TaskName		
string	Description		
date	Deadline		
string	Status		
datetime	DateSubmitted		
datetime	DateReviewed		

TASKREVIEWS			
int	ReviewID	PK	
int	OutputID	FK	
int	ReviewedBy	FK	
string	ReviewStatus		
string	ReviewNotes		
datetime	ReviewDate		

TASKOUTPUTS			
int	OutputID	PK	
int	TaskID	FK	
int	MemberID	FK	
string	OutputText		
string	FileName		
string	FilePath		
datetime	DateUploaded		
boolean	IsFinal		

Data Dictionary**Adaya IT Solution: Corporate Project Lifecycle & Workflow Management System****Users table**

Field Names	Datatype	Length	Description
UserID-PK	Int-AI	9	User's unique ID
Username	Varchar	50	Login username / email
Password	Varchar	255	Encrypted user password
Role	Varchar	30	User role (Employee, Manager, Project Manager, HR, Executive)
IsActive	Boolean	—	Account status
DateCreated	DateTime	—	Date account was created

Employees table

Field Names	Datatype	Length	Description
EmpID-PK	Int-AI	9	Employee ID
UserID-FK	Int	9	Reference to Users table
EmployeeNumber	Varchar	25	Company-assigned employee number
LastName	Varchar	30	Employee last name
FirstName	Varchar	30	Employee first name
MiddleName	Varchar	30	Employee middle name
Position	Varchar	50	Job position
Email	Varchar	50	Employee email
DateHired	Date	—	Date of employment

ProjectProposals table

Field Names	Datatype	Length	Description
ProposalID-PK	Int-AI	9	Proposal ID
EmpID-FK	Int	9	Proposing employee
Title	Varchar	50	Project title
Description	Text	255	Project description
Status	Varchar	30	Pending / Approved / Rejected / Returned
DateSubmitted	DateTime	—	Submission date

ProposalDocuments table

Field Names	Datatype	Length	Description
DocumentID-PK	Int-AI	9	Document ID
ProposalID-FK	Int	9	Related proposal
FileName	Varchar	100	Original file name
FilePath	Varchar	255	AWS S3 file location
UploadDate	DateTime	—	Date uploaded

ProjectProposalReviews table

Field Names	Datatype	Length	Description
ProposalReviewID-PK	Int-AI	9	Proposal review ID
ProposalID-FK	Int	9	Reviewed proposal
ReviewedBy-FK	Int	9	Department Head
ReviewStatus	Varchar	30	Approved / Rejected / Returned
ReviewNotes	Text	255	Feedback or reason

ReviewDate	DateTime	—	Date reviewed
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Projects table

Field Names	Datatype	Length	Description
ProjectID-PK	Int-AI	9	Project ID
ProposalID-FK	Int	9	Approved proposal
ProjectManagerID-FK	Int	9	Assigned Project Manager
ProjectName	Varchar	100	Project name
StartDate	Date	—	Project start date
EndDate	Date	—	Project end date
Status	Varchar	30	Ongoing / Completed / On-Hold

ProjectMembers table

Field Names	Datatype	Length	Description
MemberID-PK	Int-AI	9	Project member ID
ProjectID-FK	Int	9	Assigned project
EmpID-FK	Int	9	Assigned employee
Role	Varchar	30	Member role in project
DateAssigned	DateTime	—	Date assigned

Tasks table

Field Names	Datatype	Length	Description
TaskID-PK	Int-AI	9	Task ID
ProjectID-FK	Int	9	Related project

MilestoneID-FK	Int	9	Related project
MemberID-FK	Int	9	Assigned employee
TaskName	Varchar	100	Task title
Description	Text	—	Task details
Deadline	Date	—	Task deadline
Status	Varchar	30	Pending / In-Progress / Submitted / Approved / Returned
DateSubmitted	DateTime	—	Date task was submitted
DateReviewed	DateTime	—	Date reviewed by Project Manager

TaskOutputs table

Field Names	Datatype	Length	Description
OutputID-PK	Int-AI	9	Task output ID
TaskID-FK	Int	9	Related task
MemberID-FK	Int	9	Assigned project member who submitted the output
OutputText	Text	—	Direct task output input
FileName	Varchar	100	Uploaded file name
FilePath	Text	255	File storage path (AWS S3)
DateUploaded	DateTime	—	Date output was uploaded
IsFinal	Boolean	—	Marks final submission

TaskReviews table

Field Names	Datatype	Length	Description
ReviewID-PK	Int-AI	9	Task review ID

MilestoneID-FK	Int	9	Related project
MemberID-FK	Int	9	Assigned employee
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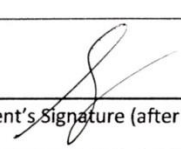
OutputID-FK	Int	9	Reviewed task output
ReviewedBy-FK	Int	9	Project Manager
ReviewStatus	Varchar	30	Approved / Returned
ReviewNotes	Varchar	— 255	Feedback
ReviewDate	DateTime	—	Review date

Milestones table

Field Names	Datatype	Length	Description
MilestoneID-PK	Int-AI	9	Milestone ID
ProjectID-FK	Int	9	Related project
Title	Varchar	100	Milestone title
TargetDate	Date	—	Target completion date
Status	Varchar	30	Pending / In-Progress / Completed
LastUpdated	DateTime	—	Auto-updated upon task approval

Notifications table

Field Names	Datatype	Length	Description
NotificationID-PK	Int-AI	9	Notification ID
UserID-FK	Int	9	Recipient
Message	Text	—	Notification message
IsRead	Boolean	—	Read status
DateSent	DateTime	—	Date sent

	ActivityLogs table			
	Field Names	Datatype	Length	Description
	LogID-PK	Int-AI	9	Log ID
	UserID-FK	Int	9	User who performed action
	Action	Varchar	100	Performed action
	Module	Varchar	50	System module
	Timestamp	DateTime	—	Date and time
Date Submitted:	February 4, 2026			
Teacher's Feedback	— Observe the data looking			
				
	Student's Signature (after feedbacking)		Teacher's Signature	

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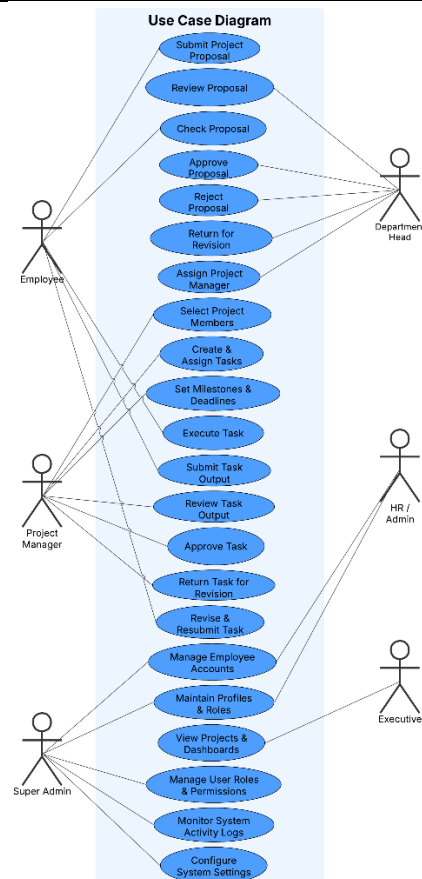
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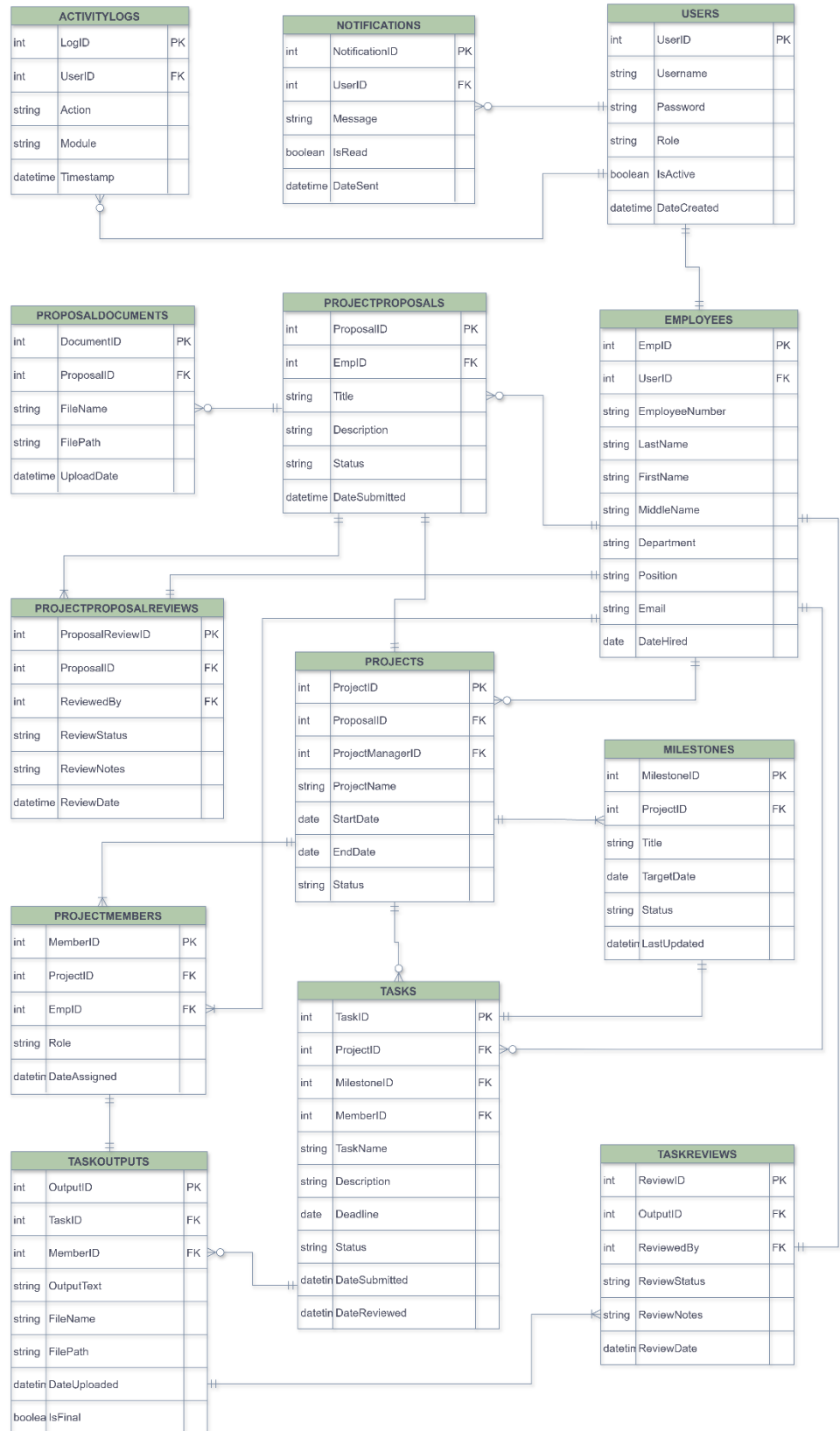
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DateCreated	DateTime	—	Date account was created

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ProposalID-FK	Int	9	Reviewed proposal
ReviewedBy-FK	Int	9	Department Head
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	ProjectID-FK	Int	9	Assigned project
	EmpID-FK	Int	9	Assigned employee
	Role	Varchar	30	Member role in project
	DateAssigned	DateTime	—	Date assigned
	Tasks table			
	Field Names	Datatype	Length	Description
	TaskID-PK	Int-AI	9	Task ID
	ProjectID-FK	Int	9	Related project

	MilestoneID-FK	Int	9	Related project
	MemberID-FK	Int	9	Assigned employee
	TaskName	Varchar	100	Task title
	Description	Varchar	255	Task details
	Deadline	Date	—	Task deadline
	Status	Varchar	30	Pending / In-Progress / Submitted / Approved / Returned
	DateSubmitted	DateTime	—	Date task was submitted
	DateReviewed	DateTime	—	Date reviewed by Project Manager
TaskOutputs table				
	Field Names	Datatype	Length	Description
	OutputID-PK	Int-AI	9	Task output ID
	TaskID-FK	Int	9	Related task
	MemberID-FK	Int	9	Assigned project member who submitted the output
	OutputText	Varchar	255	Direct task output input
	FileName	Varchar	50	Uploaded file name
	FilePath	Varchar	150	File storage path (AWS S3)
	DateUploaded	DateTime	—	Date output was uploaded
	IsFinal	Boolean	—	Marks final submission
TaskReviews table				
	Field Names	Datatype	Length	Description
	ReviewID-PK	Int-AI	9	Task review ID

OutputID-FK	Int	9	Reviewed task output
ReviewedBy-FK	Int	9	Project Manager
ReviewStatus	Varchar	30	Approved / Returned
ReviewNotes	Varchar	255	Feedback
ReviewDate	DateTime	—	Review date

Milestones table

Field Names	Datatype	Length	Description
MilestoneID-PK	Int-AI	9	Milestone ID
ProjectID-FK	Int	9	Related project
Title	Varchar	100	Milestone title
TargetDate	Date	—	Target completion date
Status	Varchar	30	Pending / In-Progress / Completed
LastUpdated	DateTime	—	Auto-updated upon task approval

Notifications table

Field Names	Datatype	Length	Description
NotificationID-PK	Int-AI	9	Notification ID
UserID-FK	Int	9	Recipient
Message	Text	—	Notification message
IsRead	Boolean	—	Read status
DateSent	DateTime	—	Date sent

	ActivityLogs table			
	Field Names	Datatype	Length	Description
	LogID-PK	Int-AI	9	Log ID
	UserID-FK	Int	9	User who performed action
	Action	Varchar	100	Performed action
	Module	Varchar	50	System module
	Timestamp	DateTime	—	Date and time
Date Submitted:	February 16, 2026			
Teacher's Feedback				
	Student's Signature (after feedbacking)Teacher's Signature			